

Solution Requirements

Date	04 July 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Users can securely access the system using a username and password (Admin login). General users can access prediction features without login.	High
2	Authorization Levels	Admin: Manage datasets, retrain models, and view reports. User: Enter flood parameters and view prediction results.	High
3	External Interfaces	Connects with weather/flood datasets and optionally weather APIs for real-time environmental data.	Medium
4	Transaction Processing	Accepts flood-related input (rainfall, water level, humidity, etc.), processes it through the ML model, and displays the prediction result.	High
5	Reporting	Generates flood prediction reports, model accuracy reports, and graphical dashboards for analysis.	Medium

S.No	Requirement Category	Requirement Description	Priority
6	Business Rules	The system validates user inputs, preprocesses the data, predicts flood risk, and displays safety recommendations based on the prediction.	High
7	Compliance to Laws or Regulations	Ensures user data privacy and follows applicable data protection guidelines. Uses datasets only for educational/research purposes.	Medium
8	Other	Allows users to save prediction history and download reports (optional feature).	Low

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The system should process flood predictions quickly.	Prediction displayed in < 2 seconds
2	Scalability	The system should support increasing users and larger datasets without performance degradation.	Supports 500+ concurrent users
3	Security & Data Privacy	Protect user data using secure authentication and encrypted communication.	HTTPS enabled and encrypted data transmission
4	Reliability & Availability	The application should remain available with minimal downtime.	99% uptime during operation
5	Usability & Accessibility	The interface should be simple, responsive, and easy to use on desktop and mobile devices.	Responsive UI with clear navigation
6	Maintainability & Portability	The system should be easy to update, maintain, and deploy on different operating systems or cloud platforms.	Modular codebase with cross-platform compatibility